

Power BI Assignment 01 – Data Transformation and Modeling

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Course Name: Program in AI-driven Data Analytics

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Introduction

This project transforms raw business data into actionable insights using Microsoft Power BI. The solution integrates three core datasets — List of Orders, Order Details, and Sales Target — to deliver a unified view of sales transactions, performance metrics, and target achievement. The resulting dashboard enables data-driven decision making through interactive visualizations of revenue, profitability, and variance analysis.

Objectives of the Project

- **Data Preparation:** Clean and standardize the List of Orders, Order Details, and Sales Target datasets using Power Query to ensure analysis-ready data
- **Metric Calculation:** Compute key business metrics including Total Amount and Total Target aggregated by Month for trend and variance analysis
- **Data Modeling:** Design a star schema data model in Power BI with defined relationships between fact and dimension tables to enable efficient querying
- **Insight Generation:** Deliver clear, interactive visualizations that highlight sales performance, profitability, and target gaps to support data-driven decision-making

Data Import & Transformation

Data Source Connection:

Established connections to the source files containing the three primary datasets: List of Orders, Order Details, and Sales Target. Verified file paths and data accessibility prior to import.

Power Query Staging:

Loaded all tables into Power Query Editor for initial profiling, cleaning, and transformation. This staging step enabled data type validation, column standardization, and preparation for subsequent merge operations.

Transformation Logic

Transformation ensures the raw data is clean, consistent, and ready for analysis. In Power Query, the following steps are typically applied:

- **Data Cleaning** → Removing duplicates, handling missing values, and correcting data types.
- **Text Formatting** → Standardizing product names, categories, and labels for consistency.

Order ID	Amount	Profit	Quantity	Category	Sub-Category
B-25601	1275	-1248	7	Furniture	Bookcases
B-25601	66	-12	5	Clothing	Stole
B-25601	8	-2	3	Clothing	Hankierchief
B-25601	80	-56	4	Electronics	Electronic Games
B-25602	168	-111	2	Electronics	Phones
B-25602	424	-272	5	Electronics	Phones
B-25602	2617	1151	4	Electronics	Phones
B-25602	561	212	3	Clothing	Saree
B-25602	119	-5	8	Clothing	Saree
B-25603	1355	-60	5	Clothing	Trousers
B-25603	24	-30	1	Furniture	Chairs
B-25603	193	-166	3	Clothing	Saree
B-25603	180	5	3	Clothing	Trousers
B-25603	116	16	4	Clothing	Stole
B-25603	107	36	6	Clothing	Stole
B-25603	12	1	2	Clothing	Hankierchief
B-25603	38	18	1	Clothing	Kurti
B-25604	65	17	2	Clothing	T-shirt
B-25604	157	5	9	Clothing	Saree
B-25605	75	0	7	Clothing	Saree
B-25606	87	4	2	Clothing	Shirt
B-25607	50	15	4	Clothing	Leggings
B-25608	1364	-1864	5	Furniture	Tables
B-25608	476	0	3	Furniture	Chairs
B-25608	257	23	5	Clothing	Hankierchief
B-25608	856	385	6	Electronics	Printers
B-25609	485	29	4	Electronics	Electronic Games
B-25609	25	-5	4	Clothing	Saree

Task:

Data Import & Transformation

1.Data Import:

- **Load the orders list:** Import List of Orders.csv into Power BI Desktop.
- **Start transforming:** After loading, use the Transform Data option to open that dataset in Power Query Editor. That's where you'll clean and shape the data.
- **Add the other files:** Also import Order Details.csv and Sales Target.csv directly into Power Query Editor.

The screenshot displays the Power Query Editor window. The main area shows a table with 5 columns: Order ID, Order Date, Customer Name, State, and City. The data is sorted by Order ID. The ribbon at the top includes tabs for File, Home, Transform, Add Column, View, Tools, and Help. The right-hand pane shows the 'Query Settings' for the 'List of Orders' query, with the 'APPLIED STEPS' list containing 'Source', 'Promoted Headers', 'Changed Type', 'Kept First Rows', and 'Data type-date'.

Order ID	Order Date	Customer Name	State	City
1	01-04-2018	Bharat	Gujarat	Ahmedabad
2	01-04-2018	Pearl	Maharashtra	Pune
3	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	03-04-2018	Divsha	Rajasthan	Jaipur
5	05-04-2018	Kasheen	West Bengal	Kolkata
6	06-04-2018	Hazel	Karnataka	Bangalore
7	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	09-04-2018	Iltesh	Uttar Pradesh	Lucknow
10	09-04-2018	Yogesh	Bihar	Patna
11	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	12-04-2018	Shrichand	Punjab	Chandigarh
13	12-04-2018	Mukesh	Haryana	Chandigarh
14	13-04-2018	Vandana	Himachal Pradesh	Simla
15	15-04-2018	Bhavna	Sikkim	Gangtok
16	15-04-2018	Kanak	Goa	Goa
17	17-04-2018	Sagar	Nagaland	Kohima
18	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	20-04-2018	Sarita	Maharashtra	Pune
21	20-04-2018	Deepak	Madhya Pradesh	Bhopal
22	22-04-2018	Monisha	Rajasthan	Jaipur
23	22-04-2018	Atharv	West Bengal	Kolkata
24	22-04-2018	Vini	Karnataka	Bangalore
25	23-04-2018	Pinky	Jammu and Kashmir	Kashmir
26	23-04-2018	Bhishm	Maharashtra	Mumbai
27	23-04-2018	Hitika	Madhya Pradesh	Indore
28	24-04-2018	Pooja	Bihar	Patna

\$ Target	\$ Amount
10,400.00	1,275.00
10,500.00	66.00
10,600.00	8.00
10,800.00	80.00
10,900.00	168.00
11,000.00	424.00
11,100.00	2,617.00
11,300.00	561.00
11,400.00	119.00
11,500.00	1,355.00
11,600.00	24.00
11,800.00	193.00
11,800.00	180.00

2. Row Limiting & Data Types:

- **Keep only 500 rows:** In Power Query, use Home > Keep Rows > Keep Top Rows and enter 500 for the List of Orders table. This is usually for testing or reducing file size while you build.
- **Fix the date column:** Select Order Date and change the data type to Date using the dropdown in the column header. Power BI often imports dates as text, and this fixes that.
- **Set money columns:** Select both Amount and Target columns and set their data type to Fixed Decimal Number. This ensures calculations like profit and totals are accurate to 2 decimal places.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Column From Custom Invoke Custom Examples * Column Function Duplicate Column

Conditional Column Index Column Merge Columns Extract Parse From Text From Number From Date & Time

Statistics Standard Scientific Rounding Information

Trigonometry Date Time Duration

Queries [3] Order Details List of Orders Sales target

Table.AddColumn(#"Removed Columns", "Location", each Text.Combine([City], [State]), "~", type text)

Order ID	Order Date	CustomerName	State	City	Location
1 B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat
2 B-25602	01-04-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra
3 B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
4 B-25604	03-04-2018	Divsha	Rajasthan	Jaipur	Jaipur,Rajasthan
5 B-25605	05-04-2018	Kasheen	West Bengal	Kolkata	Kolkata,West Bengal
6 B-25606	06-04-2018	Hazel	Karnataka	Bangalore	Bangalore,Karnataka
7 B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir
8 B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu
9 B-25609	09-04-2018	Itresh	Uttar Pradesh	Lucknow	Lucknow,Uttar Pradesh
10 B-25610	09-04-2018	Yogesh	Bihar	Patna	Patna,Bihar
11 B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram,Kerala
12 B-25612	12-04-2018	Shrichand	Punjab	Chandigarh	Chandigarh,Punjab
13 B-25613	12-04-2018	Mukesh	Haryana	Chandigarh	Chandigarh,Haryana
14 B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla	Simla,Himachal Pradesh
15 B-25615	15-04-2018	Bhavia	Sikkim	Gangtok	Gangtok,Sikkim
16 B-25616	15-04-2018	Kanak	Goa	Goa	Goa,Goa
17 B-25617	17-04-2018	Sagar	Nagaland	Kohima	Kohima,Nagaland
18 B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad	Hyderabad,Andhra Pradesh
19 B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad	Ahmedabad,Gujarat
20 B-25620	20-04-2018	Sarita	Maharashtra	Pune	Pune,Maharashtra
21 B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
22 B-25622	22-04-2018	Monisha	Rajasthan	Jaipur	Jaipur,Rajasthan
23 B-25623	22-04-2018	Atharv	West Bengal	Kolkata	Kolkata,West Bengal
24 B-25624	22-04-2018	Vini	Karnataka	Bangalore	Bangalore,Karnataka
25 B-25625	23-04-2018	Pinky	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir
26 B-25626	23-04-2018	Bhishm	Maharashtra	Mumbai	Mumbai,Maharashtra
27 B-25627	23-04-2018	Hitika	Madhya Pradesh	Indore	Indore,Madhya Pradesh
28 B-25628	24-04-2018	Pooja	Bihar	Patna	Patna,Bihar

6 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 12:01

Query Settings

PROPERTIES

Name

List of Orders

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Kept First Rows

Data type-date

Trimmed Text

Cleaned Text

Capitalized Each Word

Inserted Merged Column

Removed Columns

Inserted Merged Column1

3. Text Formatting :

- **Clean up customer names:** For the Customer Name column, go to Transform > Format > Capitalize Each Word. This converts "bharat" to "Bharath" so your reports look consistent.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Column From Custom Invoke Custom Examples * Column Function Duplicate Column

Conditional Column Index Column Merge Columns Extract Parse From Text From Number From Date & Time

Statistics Standard Scientific Rounding Information

Trigonometry Date Time Duration

Queries [3] Order Details List of Orders Sales target

Table.AddColumn(#"Removed Columns", "Location", each Text.Combine([City], [State]), "~", type text)

Order ID	Order Date	CustomerName	State	City	Location
1 B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat
2 B-25602	01-04-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra
3 B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
4 B-25604	03-04-2018	Divsha	Rajasthan	Jaipur	Jaipur,Rajasthan
5 B-25605	05-04-2018	Kasheen	West Bengal	Kolkata	Kolkata,West Bengal
6 B-25606	06-04-2018	Hazel	Karnataka	Bangalore	Bangalore,Karnataka
7 B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir
8 B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu
9 B-25609	09-04-2018	Itresh	Uttar Pradesh	Lucknow	Lucknow,Uttar Pradesh
10 B-25610	09-04-2018	Yogesh	Bihar	Patna	Patna,Bihar
11 B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram,Kerala
12 B-25612	12-04-2018	Shrichand	Punjab	Chandigarh	Chandigarh,Punjab
13 B-25613	12-04-2018	Mukesh	Haryana	Chandigarh	Chandigarh,Haryana
14 B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla	Simla,Himachal Pradesh
15 B-25615	15-04-2018	Bhavia	Sikkim	Gangtok	Gangtok,Sikkim
16 B-25616	15-04-2018	Kanak	Goa	Goa	Goa,Goa
17 B-25617	17-04-2018	Sagar	Nagaland	Kohima	Kohima,Nagaland
18 B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad	Hyderabad,Andhra Pradesh
19 B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad	Ahmedabad,Gujarat
20 B-25620	20-04-2018	Sarita	Maharashtra	Pune	Pune,Maharashtra
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22 B-25622	22-04-2018	Monisha	Rajasthan	Jaipur	Jaipur,Rajasthan
23 B-25623	22-04-2018	Atharv	West Bengal	Kolkata	Kolkata,West Bengal
24 B-25624	22-04-2018	Vini	Karnataka	Bangalore	Bangalore,Karnataka
25 B-25625	23-04-2018	Pinky	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir
26 B-25626	23-04-2018	Bhishm	Maharashtra	Mumbai	Mumbai,Maharashtra
27 B-25627	23-04-2018	Hitika	Madhya Pradesh	Indore	Indore,Madhya Pradesh
28 B-25628	24-04-2018	Pooja	Bihar	Patna	Patna,Bihar

6 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 12:01

Query Settings

PROPERTIES

Name

List of Orders

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Kept First Rows

Data type-date

Trimmed Text

Cleaned Text

Capitalized Each Word

Inserted Merged Column

Removed Columns

Inserted Merged Column1

5. Conditional Column:

- **Create a Profit Status column:** Go to Add Column > Conditional Column. Name it Profit Status.

Set the rules:

- If Profit is less than 0 → output Loss
- If Profit equals 0 → output Break-Even
- If Profit is greater than 0 → output Profit
- This converts numeric profit into text categories. Makes it easy to filter visuals by "Loss-making orders" or count how many orders were profitable.

Table: RemoveColumns(#"Expanded Order Details",{"Order Details.Order ID"})

Order ID	Order Date	CustomerName	State	City	Location	Order Data
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat	
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat	
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat	
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat	
B-25602	01-04-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra	
B-25602	01-04-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra	
B-25602	01-04-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra	
B-25602	01-04-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra	
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur	Jaipur,Rajasthan	
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur	Jaipur,Rajasthan	
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata	Kolkata,West Bengal	
B-25606	06-04-2018	Hazel	Karnataka	Bangalore	Bangalore,Karnataka	
B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir	
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
B-25609	09-04-2018	Ittesh	Uttar Pradesh	Lucknow	Lucknow,Uttar Pradesh	

6. Merging Data (Joins):

- **Combine your tables:** Use Home > Merge Queries as New . Select List of Orders as the first table and Order Details as the second.
- **Match on Order ID:** Click the Order ID column in both tables to link them. This is your common key.
- **Choose join type:** Use Left Outer if you want all orders even without details, or Inner if you only want orders that have details.

- **Name it:** After merging, expand the new column and rename the whole query to Orders Data. This becomes your main fact table.

Order ID	Order Date	CustomerName	State	City	Location
1	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat
2	01-04-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra
3	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
4	03-04-2018	Divsha	Rajasthan	Jaipur	Jaipur,Rajasthan
5	05-04-2018	Kasheen	West Bengal	Kolkata	Kolkata,West Bengal
6	06-04-2018	Hazel	Karnataka	Bangalore	Bangalore,Karnataka
7	06-04-2018	Sonalshi	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir
8	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu
9	09-04-2018	Jitesh	Uttar Pradesh	Lucknow	Lucknow,Uttar Pradesh
10	09-04-2018	Yogesh	Bihar	Patna	Patna,Bihar
11	11-04-2018	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram,Kerala
12	12-04-2018	Shrichand	Punjab	Chandigarh	Chandigarh,Punjab
13	12-04-2018	Mukesh	Haryana	Chandigarh	Chandigarh,Haryana
14	13-04-2018	Vandana	Himachal Pradesh	Simla	Simla,Himachal Pradesh
15	15-04-2018	Bhavna	Sikkim	Gangtok	Gangtok,Sikkim
16	15-04-2018	Kanak	Goa	Goa	Goa,Goa
17	17-04-2018	Sagar	Nagaland	Kohima	Kohima,Nagaland
18	18-04-2018	Manju	Andhra Pradesh	Hyderabad	Hyderabad,Andhra Pradesh
19	18-04-2018	Ramesh	Gujarat	Ahmedabad	Ahmedabad,Gujarat
20	20-04-2018	Sarita	Maharashtra	Pune	Pune,Maharashtra
21	20-04-2018	Deepak	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh
22	22-04-2018	Monisha	Rajasthan	Jaipur	Jaipur,Rajasthan
23	22-04-2018	Atharv	West Bengal	Kolkata	Kolkata,West Bengal
24	22-04-2018	Vini	Karnataka	Bangalore	Bangalore,Karnataka
25	23-04-2018	Pinky	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir
26	23-04-2018	Bhishm	Maharashtra	Mumbai	Mumbai,Maharashtra
27	23-04-2018	Hitika	Madhya Pradesh	Indore	Indore,Madhya Pradesh

7. Handling Missing Data & Duplicate Data :

- **Find the gaps:** In each table, use Home > Replace Values or check column quality via View > Column Quality. Look for null or blank cells.

- **Fix strategies:**

Replace missing values: For numbers, use 0. For text, use "Unknown" or "N/A". Use Transform > Replace Values.

Remove duplicates: Select key columns like Order ID + Product ID, then Home > Remove Rows > Remove Duplicates.

- Missing data breaks calculations and visuals. Duplicates inflate totals. Cleaning this now prevents headaches in your final dashboard.

Order ID	Order Date	CustomerName	State	City	Location	Order Date
1	31-03-2019	Hitika	Madhya Pradesh	Indore	Indore,Madhya Pradesh	
2	31-03-2019	Hitika	Madhya Pradesh	Indore	Indore,Madhya Pradesh	
3	31-03-2019	Hitika	Madhya Pradesh	Indore	Indore,Madhya Pradesh	
4	30-03-2019	Bhishm	Maharashtra	Mumbai	Mumbai,Maharashtra	
5	30-03-2019	Bhishm	Maharashtra	Mumbai	Mumbai,Maharashtra	
6	30-03-2019	Bhishm	Maharashtra	Mumbai	Mumbai,Maharashtra	
7	30-03-2019	Bhishm	Maharashtra	Mumbai	Mumbai,Maharashtra	
8	29-03-2019	Pinky	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir	
9	29-03-2019	Pinky	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir	
10	29-03-2019	Pinky	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir	
11	29-03-2019	Pinky	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir	
12	29-03-2019	Pinky	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir	
13	29-03-2019	Pinky	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir	
14	28-03-2019	Atharv	West Bengal	Kolkata	Kolkata,West Bengal	
15	28-03-2019	Vini	Karnataka	Bangalore	Bangalore,Karnataka	
16	28-03-2019	Vini	Karnataka	Bangalore	Bangalore,Karnataka	
17	28-03-2019	Atharv	West Bengal	Kolkata	Kolkata,West Bengal	
18	28-03-2019	Vini	Karnataka	Bangalore	Bangalore,Karnataka	
19	28-03-2019	Atharv	West Bengal	Kolkata	Kolkata,West Bengal	
20	28-03-2019	Atharv	West Bengal	Kolkata	Kolkata,West Bengal	
21	28-03-2019	Vini	Karnataka	Bangalore	Bangalore,Karnataka	
22	28-03-2019	Vini	Karnataka	Bangalore	Bangalore,Karnataka	
23	28-03-2019	Vini	Karnataka	Bangalore	Bangalore,Karnataka	
24	28-03-2019	Monisha	Rajasthan	Jaipur	Jaipur,Rajasthan	
25	28-03-2019	Atharv	West Bengal	Kolkata	Kolkata,West Bengal	
26	28-03-2019	Atharv	West Bengal	Kolkata	Kolkata,West Bengal	
27	28-03-2019	Vini	Karnataka	Bangalore	Bangalore,Karnataka	

Table: SelectRows(#"Sorted Rows", each ([State] = "Tamil Nadu"))

Order ID	Order Date	CustomerName	State	City	Location	Order Details
1	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
2	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
3	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
4	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
5	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
6	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
7	14-02-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
8	14-02-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
9	09-02-2019	Kalyani	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
10	09-02-2019	Kalyani	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
11	09-02-2019	Kalyani	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
12	09-02-2019	Kalyani	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
13	15-11-2018	Akshay	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
14	21-09-2018	Dinesh	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
15	11-07-2018	Surabhi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
16	23-06-2018	Amisha	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
17	23-06-2018	Amisha	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
18	23-06-2018	Amisha	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
19	23-06-2018	Amisha	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
20	23-06-2018	Amisha	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
21	23-06-2018	Amisha	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
22	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
23	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
24	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	
25	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	

8.Sorting and Filtering Data:

- **Sort by Order Date:** In your Orders Data table, click the dropdown arrow on the Order Date column header. Select Sort Descending.
- **Why descending:** Puts the newest orders at the top. Helps you quickly check recent data and makes sense for time-based analysis.
- **Filter by State:** Click the dropdown on your State column. Uncheck Select All, then check only Tamil Nadu.
- **Why filter:** This isn't permanent. It's for testing. You're checking if your state-level data looks right before building the full report. You can remove the filter later. Pro tip: Since you're in Chennai, Tamil Nadu, this lets you validate with local data you understand.

File Home Transform Add Column

Standard Scientific Rounding Information

Queries [4]

Order Details 1500

List of Orders

Sales target

Orders Data

Order ID	Count
B-25601	4
B-25602	5
B-25603	8
B-25604	2
B-25605	1
B-25606	1
B-25607	1
B-25608	4
B-25609	2
B-25610	6
B-25611	1
B-25612	1
B-25613	1
B-25614	2
B-25615	1
B-25616	4
B-25617	1
B-25618	2
B-25619	1
B-25620	1
B-25621	3
B-25622	1

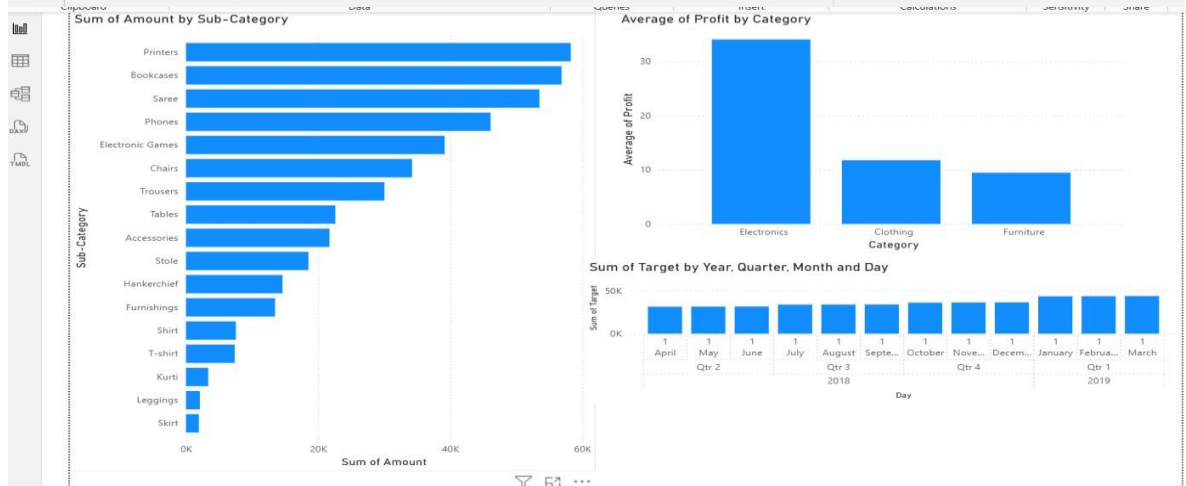
11.762908324552161 34.071428571428569 9.4567901234568126

Any Column

`= Table.RenameColumns("#Grouped Rows",{"Count"`

Category	Average
Furniture	9.456790123
Clothing	11.76290832
Electronics	34.07142857

Sub-Category	Total Amount
Bookcases	56861
Stole	18546
Hankkerchief	14608
Electronic Games	39168
Phones	46119
Saree	53511
Trousers	30039
Chairs	34222
Kurti	3361
T-shirt	7382
Shirt	7555
Leggings	2106
Tables	22614
Printers	58252
Accessories	21728
Furnishings	13484
Skirt	1946



9. Grouping and Aggregating Data:

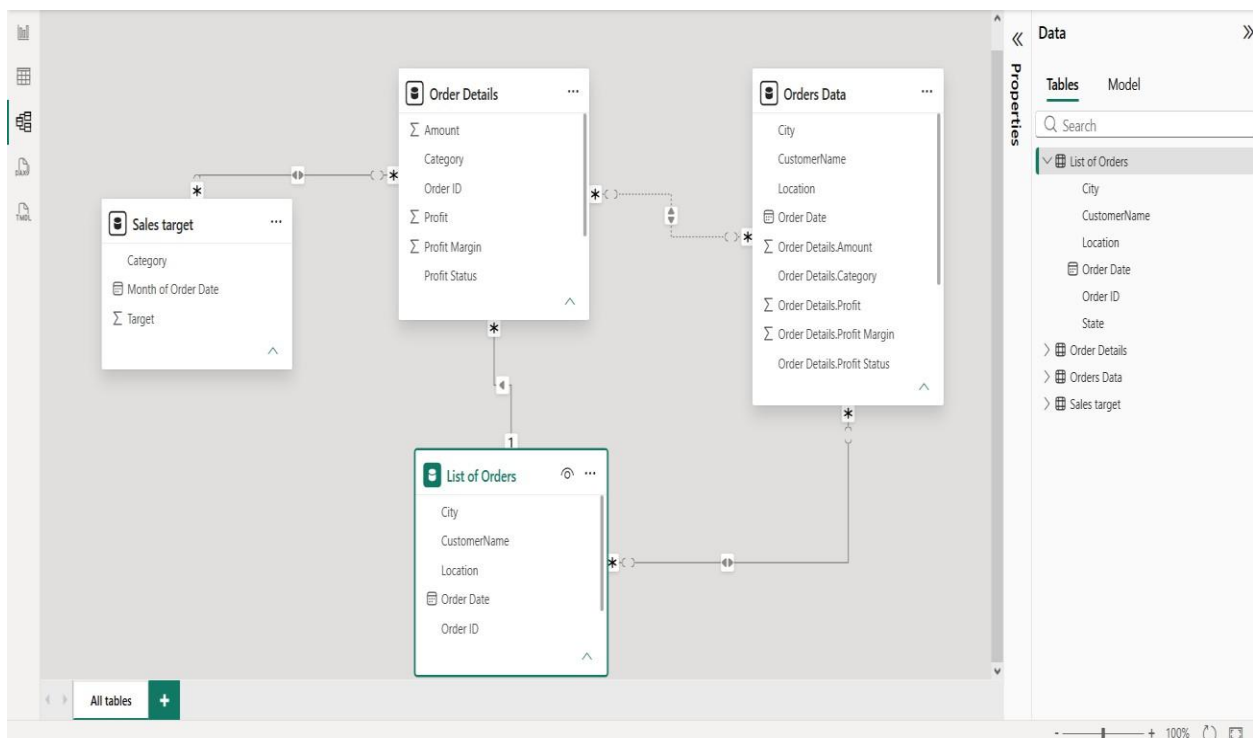
- **Duplicate tables first:** Always duplicate before you group. Grouping deletes rows. Right-click Order Details > Duplicate. Rename it Order Details Summary. Do the same for Sales Target and call it Sales Target Summary.
- **On Order Details Summary do 3 aggregations:** Use Home > Group By.

Count of Order ID: Choose Order ID and operation Count Rows. This tells you how many line items each order has, or total orders if grouped differently.

Average Profit by Category: Group by Category, then add aggregation Average of Profit. This shows which product categories are most profitable on average.

Total Amount by Sub-Category: Group by Sub-Category, then add aggregation Sum of Amount. This gives total sales per sub-category like "Chairs" or "Phones".

- **On Sales Target Summary:** Group by Month and Sum the Target column. Now you have monthly targets to compare against actual monthly sales.



10.Data Modeling:

1. Create the key relationships

List of Orders ↔ Order Details using Order ID:

- Drag Order ID from List of Orders to Order ID in Order Details.
- This should be One-to-Many. One order in List of Orders can have many line items in Order Details.
- Make sure the arrow points from List of Orders to Order Details.
- Why: Lets you slice total sales from Order Details by Order Date or State from List of Orders.

Order Details ↔ Sales Target using Category:

- Drag Category from Order Details to Category in Sales Target.
- This should also be Many-to-One if each category has one target row per month.
- Why: Lets you build "Actual Sales vs Target" charts by Category. If your targets are monthly, you may need to relate on Category + Month instead. **2.**

Use Manage Relationships to double-check Go to Modeling > Manage Relationships.

Check these 3 things:

- **Active:** Only one relationship between two tables can be active. The solid line means active.
- **Cardinality:** Should say One to Many (1:*) for Orders to Details. If it says Many to Many, you likely have duplicate Order ID values. Fix that in Power Query first.
- **Cross filter direction:** Single is usually safest. It means filters flow from the "one" side to the "many" sides only.

Conclusion

This project demonstrates the end-to-end process of converting raw business data into actionable insights using Power Query and Power BI. Through systematic ETL operations including data cleaning, text formatting, calculated column creation,

conditional logic, grouping, and table merges, the datasets were standardized for accurate analysis.

The star schema data model built from List of Orders, Order Details, and Sales Target enabled reliable comparisons of actual sales versus targets across categories and time periods. Establishing proper relationships ensured data integrity and consistent reporting logic.

Overall, this project reinforces that data preparation and modeling are foundational to business intelligence. Clean, well-structured data combined with a robust model delivers dashboards that drive informed, confident decision-making.

Thank You
Entri Elevate